The role of AWS Rekognition, IoT Core and Greengrass

IoT based beacons and cameras can be installed throughout the airport terminal to detect passenger movements. AWS Greengrass can be used to deploy AWS Lambda functions that capture images from local cameras/sensors and send them to AWS Rekognition for analysis. The camera can capture images or videos at regular intervals (5/10 seconds) and send them to AWS IoT Gateway.

AWS Rekognition's face detection ability can be used to analyse the images or videos, detect the presence of individual passengers, and track their movements. Based on the analysis results, alerts can be generated, and notifications can be pushed if there is a high passenger density or congestion in a specific area, or if a passenger is moving to a wrong gate.

The analysis results can be sent back to AWS Greengrass for further processing or action, such as triggering local alarms or notifications.

The making of an AI/ML-based flight boarding solution

Algorithm selection

Our solution uses the XGBoost algorithm for binary classification, a popular choice for predicting an event's occurrence based on a set of input features. We are trying to predict if a checked-in passenger who is shopping will reach the boarding gate 30 minutes prior to the flight's departure time.

The built-in XGBoost algorithm in SageMaker makes it easy to train and deploy powerful machine learning models. With a little bit of data preparation and some tuning of hyperparameters, building models quickly to make accurate predictions on a variety of tasks is a straightforward process.

SageMaker provisions Jupyter Notebook in the cloud, which is easy to create and use. Here are the steps.

Prepare data: Our data should be in a format that XGBoost can work with. This typically means a CSV file with columns for features and a target column for the label.

Upload data: We need to upload data to an S3 bucket so that SageMaker can access it.

Create a training job: We can create a training job by specifying the location of our training data in S3/ Datalake and the hyperparameters we want to use for the XGBoost algorithm. This can be done through the SageMaker console, the SageMaker SDK, or the AWS CLI.

Monitor the training job: Once the training job is started, we can monitor its progress through the SageMaker console or the SDK.

Deploy the model: After the training job is complete, we can deploy the trained model as an endpoint that can be used for inference.

Test the model: We can test the deployed model by sending new data and observing the predictions it produces.

Input data

The input data for our model consists of a set of features related to the passenger's movement within the airport, their shopping behaviour, and their time of arrival at the airport. These features are collected using IoT devices such as cameras, beacons, and sensors placed throughout the airport.

The following are the input features we used in our model.

Time of arrival: The data on the time at which the passenger arrives at the airport is collected from the passengers who have already done a web check-in. The IoT sensors at the entrance of the airport collect this data and pass it to the backend data lake. For non-web checked-in passengers, this is collected from the counter or the kiosk from where the boarding pass is obtained.

Passenger state: This indicates if the passenger has cleared immigration, security, checked-in, etc.

Shopping duration: This is the amount of time passengers spend shopping after getting their boarding pass and immigration/security clearance.

Distance from gate: This is the distance between the shopping area and the boarding gate, and is calculated using the IoT sensors placed throughout the duty-free shopping area.

Gate arrival time: This is the time at which the passenger arrives at the boarding gate. The prediction time will keep on getting calculated and updated in another table, till the passenger reaches the boarding gate. Continuous push message alerts will be sent to the passenger's mobile phone/smart device and also to the shopkeeper in the proximity of the passenger. At the right time, the shopkeeper's help will be sought to alert the passenger to move towards the gate to board the flight, automatically. The TV display will also alert the passenger to move towards the gate. The passenger will also be tagged on social media and asked to move towards the boarding gate. All these things will happen automatically thanks to various AWS AI/ML, IoT and other services.

Scheduled departure time: This is the scheduled departure time of the passenger's flight.

Flight delay: This indicates any delay in the scheduled departure time of the passenger's flight.

Walking pattern: This is the walking pattern captured from the passenger's wearable device or smartphone. If not available, the default value will be used.

Hand luggage: This data will be collected at the time of check-in but may become inconsistent due to shopping add-ons.

Terminal congestion: This is a measure of how busy the terminal is at the time the passenger arrives at the airport, and on the path to the gate. A heat map will also be displayed on the passenger's smart device.